

Exercise 1

Problem 1 (Simple calculations on multi-layer perceptrons):

Consider a L -layer multi-layer perceptron (MLP) without bias, with input $\mathbf{x} \in \mathbb{R}^d$, outputs $\mathbf{y} \in \mathbb{R}^k$, with a hidden layer of width w for $L \geq 2$. This MLP consists thus of layers $\mathbf{W}_1 \in \mathbb{R}^{w \times d}, \dots, \mathbf{W}_L \in \mathbb{R}^{k \times w}$, and $\mathbf{W}_i \in \mathbb{R}^{w \times w}$ for $1 < i < L$.

- What is the total number of parameters of the MLP and how does it scale with the depth and the width of the model?
- How many scalar addition and multiplication operations are needed for a single forward pass?
Hint: Consider first what the complexity of a single matrix-matrix or matrix-vector multiplication of some given dimensions is.
- How does the computational complexity of a single forward pass scale with depth L and width w ?
- (Bonus) What is the computational complexity of a single backward pass via backpropagation?

a) for $L=2$, we have $\mathbf{W}_1 \in \mathbb{R}^{w \times d}$ and $\mathbf{W}_2 \in \mathbb{R}^{k \times w}$, so we have $w(k+d)$ param.

more generally, an L -layer MLP w/ $L \geq 2$ has

$w \cdot ((L-2)w + k + d)$ parameters.

b) Our MLP will have the form:

output $= f_L(f_{L-1}(\dots f_1(\mathbf{x}) \dots))$, where each $\mathbf{a}^l = f_l(\mathbf{x}) = \sigma(\mathbf{W}^l \mathbf{a}^{l-1} + \mathbf{b}^l)$

we ignore the bias term here, which can get absorbed into $\mathbf{W}$

w/ some non-linearity $\sigma(\cdot)$, which is applied element-wise.

For a general matrix-vector multiplication $\mathbf{M} \cdot \mathbf{v}$, with $\mathbf{M} \in \mathbb{R}^{m \times n}$, $\mathbf{v} \in \mathbb{R}^n$
we compute each entry via $(\mathbf{M} \cdot \mathbf{v})_j = \sum_{i=1}^n \mathbf{M}_{ij} \cdot \mathbf{v}_j$, so we perform

n scalar mult. & additions per entry

or $m \cdot n$ for the full Mat-mul.

In the MLP we have one Matmul w/ $\mathbf{W}_1 \in \mathbb{R}^{w \times d}$, $\mathbf{x} \in \mathbb{R}^d$
one Matmul w/ $\mathbf{W}_L \in \mathbb{R}^{k \times w}$, $\mathbf{a}_L \in \mathbb{R}^w$
and $L-2$ Matmults w/ $\mathbf{W}_i \in \mathbb{R}^{w \times w}$, $\mathbf{a}_i \in \mathbb{R}^w$
so $w \cdot d + k \cdot w + (L-2)w^2$ scalar mult. + add.

c) The forward pass has a complexity of $O(L \cdot w^2)$.

Problem 2 (Sigmoid vs. Hyperbolic Tangent):

For any activation function $\sigma : \mathbb{R} \rightarrow \mathbb{R}$, we extend its definition to multivariate case by entry-wise evaluation:

$$\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n, (x_i)_{i=1}^n \mapsto (\sigma(x_i))_{i=1}^n.$$

Consider the following two-layer feedforward network with $\left(\tanh : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \frac{e^x - e^{-x}}{e^x + e^{-x}} \right)$ as the activation function:

$$F : \mathbb{R}^d \rightarrow \mathbb{R}^m, \mathbf{x} \mapsto \mathbf{W}_2 \tanh(\mathbf{W}_1 \mathbf{x} + b_1) + b_2,$$

where $\mathbf{x} \in \mathbb{R}^d$, $\mathbf{W}_1 \in \mathbb{R}^{n \times d}$, $b_1 \in \mathbb{R}^n$, $\mathbf{W}_2 \in \mathbb{R}^{m \times n}$, $b_2 \in \mathbb{R}^m$.

a) Show that there exists a two-layer network

$$G(\mathbf{x}) = \mathbf{W}'_2 s_1(\mathbf{W}'_1 \mathbf{x} + b'_1) + b'_2,$$

which computes exactly the same function as $F(x)$ using weight matrices and bias vectors of the same dimension, but with the sigmoid activation function s_1 . Write down the weights $\mathbf{W}'_1, \mathbf{W}'_2, b'_1, b'_2$ in terms of $\mathbf{W}_1, \mathbf{W}_2, b_1, b_2$.

(Hint: You can first show that $\tanh(x) = 2s_1(2x) - 1$, where s_1 is the Sigmoid function.)

b) Does the same statement hold if s_1 is replaced by l_1 ? Why?

$$a) F(x) = W_2 \tanh(W_1 x + b_1) + b_2$$

$$\text{Note: } \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$s_1(x) = \frac{1}{1+e^{-x}} \Rightarrow \text{Using hint:}$$

$$\begin{aligned} 2s_1(2x) - 1 &= \frac{2}{1+e^{-2x}} - \frac{1+e^{-2x}}{1+e^{-2x}} \\ &= \frac{1-e^{-2x}}{1+e^{-2x}} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \tanh(x) \end{aligned}$$

~ rewrite tanh using s_1 & reshape:

$$W_2 \tanh(W_1 x + b_1) + b_2 = W_2 \cdot [2s_1(2(W_1 x + b_1)) - \mathbf{1}] + b_2$$

$$= \underbrace{2W_2}_{W'_2} s_1(\underbrace{2W_1}_{W'_1} x + \underbrace{2b_1}_{b'_1}) - \underbrace{W_2 \mathbf{1}}_{b'_2} + b_2$$

vector of ones $\mathbf{1} \in \mathbb{R}^n$

b) No. $\tanh$ is continuous $\rightarrow F(x)$ is continuous,
but l_1 is not $\Rightarrow G(x)$ would not be
continuous for any W'_2, W'_1, b'_2, b'_1

Problem 3 (Gradient for Three Layer network with softmax activation and cross-entropy loss):

Consider a three layer neural network with a softmax activation at the output defined as

$$f_{\mathbf{W}, \Theta}(\mathbf{x}) = \sigma^{\max}(\mathbf{W}\mathbf{x}; \Theta),$$

where the softmax function converts the logits $\mathbf{W}\mathbf{x}$ into a probability distribution via

$$\sigma_i^{\max}(\mathbf{x}; \Theta) = \frac{\exp(\mathbf{x}^\top \theta_i)}{\sum_{j=1}^k \exp(\mathbf{x}^\top \theta_j)}, \quad \Theta = [\theta_1, \dots, \theta_k]. \quad \left. \vphantom{\sum_{j=1}^k} \right\} \sum_{i=1}^k \sigma_i^{\max}(\mathbf{x}; \Theta) = 1$$

The cross-entropy loss is chosen to train the network, i.e.

$$\ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta) = -\mathbf{y}^\top \ln(\sigma^{\max}(\mathbf{W}\mathbf{x}; \Theta)), \quad \text{why is } \ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta) \geq 0? \\ \text{why } \ell > 0?$$

a) Derive the derivative with respect to each θ_i , i.e. $\frac{\partial \ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta)}{\partial \theta_i}$.

b) Derive the derivative with respect to $\mathbf{W}$, i.e. $\frac{\partial \ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta)}{\partial \mathbf{W}}$.

Hint: Make use of the matrix calculus to simplify the computation. Alternatively, calculate the the derivative entry-wise and combine it into a matrix expression.

a) compute $\frac{\partial \ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta)}{\partial \theta_i}$:

$$\ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta) = -\mathbf{y}^\top \ln(\sigma^{\max}(\mathbf{W}\mathbf{x}; \Theta))$$

$$= -\mathbf{y}^\top \ln \begin{pmatrix} \vdots \\ \frac{\exp(\theta_i^\top (\mathbf{W}\mathbf{x}))}{\sum_{j=1}^k \exp(\theta_j^\top (\mathbf{W}\mathbf{x}))} \\ \vdots \end{pmatrix}$$

$$= -\sum_{\ell=1}^k y_\ell \cdot \ln \left(\frac{\exp(\theta_\ell^\top (\mathbf{W}\mathbf{x}))}{\sum_{j=1}^k \exp(\theta_j^\top (\mathbf{W}\mathbf{x}))} \right) = -\sum_{\ell=1}^k y_\ell \cdot (\theta_\ell^\top (\mathbf{W}\mathbf{x})) \\ + \left(\sum_{\ell=1}^k y_\ell \right) \ln \left(\sum_{j=1}^k \exp(\theta_j^\top (\mathbf{W}\mathbf{x})) \right)$$

$$\frac{\partial}{\partial \theta_i} \ell(\mathbf{x}, \mathbf{y}; \mathbf{W}, \Theta) = -y_i \cdot \mathbf{W}\mathbf{x} + \left(\sum_{\ell=1}^k y_\ell \right) \cdot \frac{1}{\sum_{j=1}^k \exp(\theta_j^\top (\mathbf{W}\mathbf{x}))} \cdot \exp(\theta_i^\top (\mathbf{W}\mathbf{x})) \cdot \mathbf{W}\mathbf{x}$$

$$= -y_i \cdot \mathbf{W}\mathbf{x} + \left(\sum_{\ell=1}^k y_\ell \right) \cdot \frac{\exp(\theta_i^\top (\mathbf{W}\mathbf{x}))}{\sum_{j=1}^k \exp(\theta_j^\top (\mathbf{W}\mathbf{x}))} \cdot \mathbf{W}\mathbf{x}$$

$$= (-y_i + \underbrace{\left(\sum_{\ell=1}^k y_{\ell}\right)}_{\substack{\text{if } y \text{ is} \\ \text{normalized} \\ \text{or one-hot-vector}}} \cdot \sigma_i^{\max}(Wx; \Theta)) \cdot Wx = (-y_i + \sigma_i^{\max}(Wx; \Theta)) \cdot Wx$$

if y is
one-hot
vector

$\Rightarrow \sum_{\ell=1}^k y_{\ell} = 1$

b) compute $\frac{\partial}{\partial W} \ell(x, y; W, \Theta)$

$$\frac{\partial}{\partial W} \left(-\sum_{\ell=1}^k y_{\ell} \cdot (\theta_{\ell}^T Wx) + \left(\sum_{\ell=1}^k y_{\ell} \right) \ln \left(\sum_{j=1}^k \underbrace{\exp(\theta_j^T Wx)}_{=: g_j(W)} \right) \right)$$

$\frac{\partial}{\partial W} g_j(W) = \exp(\theta_j^T Wx) \cdot \theta_j x^T$

$$= -\sum_{\ell=1}^k y_{\ell} \cdot \theta_{\ell} x^T + \left(\sum_{\ell=1}^k y_{\ell} \right) \cdot \underbrace{\frac{\partial}{\partial W} \left(\ln \left(\sum_{j=1}^k g_j(W) \right) \right)}_{= \frac{1}{\sum_{j=1}^k g_j(W)} \cdot \left(\sum_{j=1}^k \frac{\partial}{\partial W} g_j(W) \right)}$$

matrix cookbook Eq. (70)

$$\frac{\partial a^T X b}{\partial X} = \underline{a} \underline{b}^T$$

$$= -\sum_{\ell=1}^k y_{\ell} \cdot \theta_{\ell} x^T + \left(\sum_{m=1}^k y_m \right) \cdot \frac{1}{\sum_{j=1}^k \exp(\theta_j^T Wx)} \cdot \left(\sum_{\ell=1}^k \exp(\theta_{\ell}^T Wx) \cdot \theta_{\ell} x^T \right)$$

$$= -\sum_{\ell=1}^k y_{\ell} \cdot \theta_{\ell} x^T + \left(\sum_{m=1}^k y_m \right) \cdot \underbrace{\sum_{\ell=1}^k \frac{\exp(\theta_{\ell}^T Wx)}{\sum_{j=1}^k \exp(\theta_j^T Wx)}}_{\sigma_i^{\max}(Wx; \Theta)} \cdot \theta_{\ell} x^T$$

$$= -\sum_{\ell=1}^k \left(y_{\ell} - \left(\sum_{j=1}^k y_j \right) \sigma_i^{\max}(Wx; \Theta) \right) \cdot \theta_{\ell} x^T$$

$$\stackrel{\uparrow}{=} -\sum_{\ell=1}^k \left(y_{\ell} - \sigma_i^{\max}(Wx; \Theta) \right) \cdot \theta_{\ell} x^T$$

if y is
one-hot-vector